Generating Personalized Games with Sensing Data to Support Self-Reported Data Collection

Jin-seo Kim
Northeastern University
Boston, Massachusetts, USA
kim.jinseo@northeastern.edu

Ha Le
Northeastern University
Boston, Massachusetts, USA
le.ha1@northeastern.edu

Akshat Choube
Northeastern University
Boston, Massachusetts, USA
choube.a@northeastern.edu

Varun Mishra
Northeastern University
Boston, Massachusetts, USA
v.mishra@northeastern.edu

Stephen S Intille
Northeastern University
Boston, Massachusetts, USA
s.intille@northeastern.edu

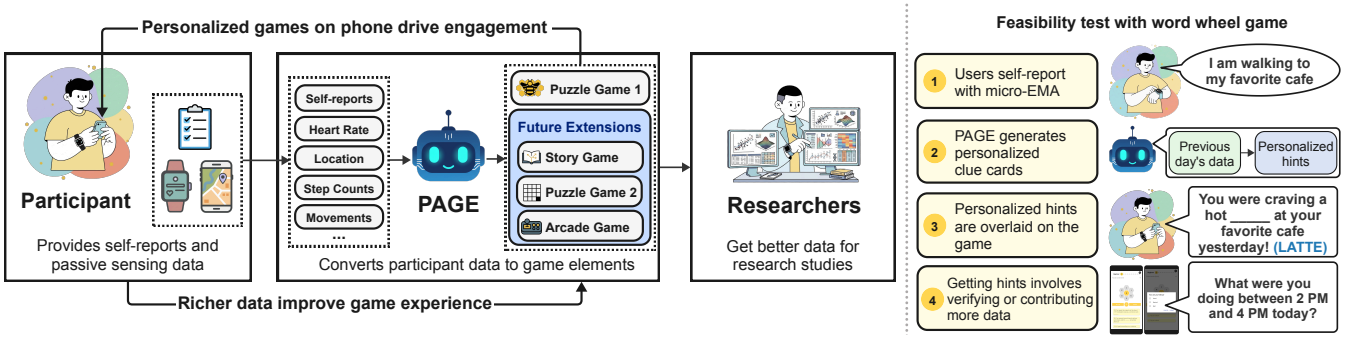


Figure 1: PAGE is an approach for converting participants' self-reports and passive sensing data into personalized games (left) aimed at creating a positive feedback loop: personalized games drive engagement, incentivizing users to provide richer data, which further improves the game and yields richer data. We tested the feasibility of PAGE using a word wheel game (right): (1) participants self-reported their activities via smartwatch; (2) our system generated personalized puzzle clues from the data; and (3) participants played the personalized game on their smartphones.

Abstract

Collecting in-situ behavioral data to support research studies often involves prompting repeated ecological momentary assessments (EMAs) on smartphones and wearables. Sustained engagement, however, remains a major challenge because study participants often experience survey fatigue and declining motivation over time. To address this challenge, we introduce Personalized Adaptive Games for Engagement (PAGE): a novel LLM-powered approach that integrates data collection with a personalized game experience. PAGE uses a participant's previously collected data to dynamically generate personalized puzzles, embedding behavioral data collection within engaging gameplay. We deployed a prototype using a variant of a word wheel puzzle game in a two-week feasibility study (N=15); while participants responded to EMAs in both weeks of the study, they also received sensor-based personalized puzzles in one of those weeks. We found that most participants (80%) preferred the week with personalized puzzles. Participants also contributed richer EMA data (72 additional unique contextual tags) and 1,188 additional uncompensated responses via a feasible in-game data collection channel during the puzzle week. We discuss design guidelines for developing context-aware personalized games to support engagement, potentially for use in longitudinal studies.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Ubiquitous and mobile computing systems and tools**.

Keywords

Data Collection, Mobile and Wearable Sensing, Personalization, Gamification, Personalized Games, User Engagement, AI and LLM

ACM Reference Format:

Jin-seo Kim, Ha Le, Akshat Choube, Varun Mishra, and Stephen S Intille. 2026. Generating Personalized Games with Sensing Data to Support Self-Reported Data Collection. In *The 39th Annual ACM Symposium on User Interface Software and Technology (UIST '26), November 02–05, 2026, Detroit, MI, USA.* ACM, New York, NY, USA, 21 pages. https://doi.org/10.1145/3830398.3830678

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UIST '26, Detroit, MI, USA

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ACM ISBN 979-8-4007-2856-3/26/11
https://doi.org/10.1145/3830398.3830678

1 Introduction

Collecting in-situ data on participants' behaviors, states, and contexts in research studies is crucial in fields such as human-computer interaction, medicine, public health, and behavioral science. In the past, researchers have used such data to understand health outcomes [3, 44, 96, 97], study mental health and wellbeing [16, 17, 27, 85], study how diseases spread [59, 83], train predictive models and algorithms [75], and design interventions [12, 25]. To better capture how individuals change over time and to identify behavioral trends in large populations, researchers often run *longitudinal* data collection studies. In these studies, repeated measurements are taken from participants over long periods, often spanning weeks, months, or even years [6].

In longitudinal data collection studies, researchers collect data consisting of both passive sensing data and self-reports [65]. *Passive sensing data* can be collected using sensors present in participants' smartphones and wearable devices, or through additional devices that may be worn or installed (e.g., accelerometers, beacons, and smart speakers) [11, 65]. To complement passive sensing data, researchers also collect *self-reports* from participants, typically through surveys and questionnaires delivered on participants' mobile and wearable devices. Self-reports (1) help researchers verify and make sense of the passive sensing data [13, 21, 35], (2) add context to missing sensing data, and (3) measure behaviors and states that cannot be passively determined (e.g., stress, happiness) [1, 35, 55, 56].

Longitudinal data collection studies, however, often suffer from participants dropping out and losing interest over time. When studies run for weeks, months, or years, people can get tired of repeated measurements and lose motivation to continue participating [31, 76]. In short-term studies, researchers often provide financial compensation for data collection and for answering momentary self-reports. For extended periods and large participant pools, this approach can become expensive and impractical [91].

Increasingly, researchers are interested in developing *massive* health surveillance studies with volunteers willing to contribute data longitudinally. An example is the All of Us Research Program, led by the U.S. National Institutes of Health, which had enrolled over 873k participants as of early 2026 [14, 67, 68]. Participants volunteer with *no monetary compensation*, remain in the study indefinitely, and can opt-in to contribute continuous data from their consumer FitBit wearable, if they wear one. Due to concerns about participant burden leading to study attrition, participants are rarely asked to contribute self-report data—typically 2-3 short surveys a year [58, 77]. Unlike smaller studies, the massive scale of the study makes it impractical to financially compensate participants for providing data on a regular basis. The sparsity at which self-reports are being collected, and the complete lack of momentary self-reports, is a missed opportunity to understand dynamic behavioral patterns; rich, longitudinal passive and self-report data from a subset of the study population could enable important new discoveries.

In small EMA studies, researchers have explored adding game-like elements (e.g., points, leaderboards, social competition) to their data collection systems as one way to motivate and sustain cost-efficient and burden-acceptable self-reports [38, 53, 86]. Such approaches, however, only generically overlay gamified features on the self-report experience; they do not explicitly tailor gameplay to the behavioral information collected via passive sensing or self-reports. Moreover, they do not use the game features to improve the type and quality of data obtained.

We introduce a novel approach that can augment popular word games with the goal of transforming them into engagement-driving experiences for self-reported data collection. Our approach combines the genuine, intrinsic motivation that some participants may have to play and enjoy puzzle games with data-collection methods. The games adapt based on collected data for a more personalized experience, encouraging participants to share *additional and richer data*. We explore three research questions:

- **RQ1.** How might personalized, adaptive games affect participant engagement in data collection studies?
- **RQ2.** How might participants interact with personalized games, and what are the characteristics of the additional data collected when using such a game?
- **RQ3.** What are participants' attitudes and acceptance towards adaptive games used to collect behavioral data, and what design opportunities exist to further enhance participant experience?

To address these questions, we built a proof-of-concept (PoC) system and used it to run a two-week feasibility study with 15 participants. The key **contributions** of our work are:

- We introduce Personalized Adaptive Games for Engagement (PAGE), a novel approach for using an LLM to augment word puzzle games to create a personalized game experience that is tailored with passively sensed and self-reported data; PAGE is intended to help support participant engagement and the richness of self-reported data acquired in behavioral measurement studies. We implemented a PoC system based on our approach using a variant of a word wheel puzzle.
- We conducted a mixed-method, within-subjects user study with 15 participants over two weeks during which a system based on PAGE was used to collect self-reports on participants' physical activity and postures; our aim was to explore the feasibility and usability of the system.
- We found that participants preferred the data collection experience with personalized games over the same protocol without the games, and contributed richer, more diverse self-reports when they engaged with the game.
- Based on observations from the feasibility study, we suggest design guidelines researchers might consider to create adaptive games that might be used to support engagement in future longitudinal studies.

2 Related work

Our proposed system design builds upon prior work on collecting activity and behavioral data via EMA in longitudinal research studies, the increased use of LLMs to make sense of passive sensing data, and how gamification can potentially increase participants' engagement and improve data quality.

2.1 Collecting behavioral data in longitudinal studies

Collecting human activity and behavioral data over time is a central goal in HCI, health informatics, and behavioral science. Large-scale studies increasingly aim to capture longitudinal data to understand how behaviors evolve and to support personalized interventions [2, 5, 29]. Collecting high-resolution data over extended periods, however, remains challenging. Participants often experience burden and fatigue, leading to declining engagement and incomplete datasets. As a result, many studies rely on financial incentives to sustain participation [89], raising concerns about scalability and long-term sustainability.

Ecological momentary assessment (EMA) is a widely used approach for collecting in-situ behavioral data by prompting participants throughout the day via mobile or wearable devices [10, 78]. While EMA enables fine-grained measurement, it is inherently interruptive and repetitive, often resulting in missing data, contextual response bias, and reduced engagement over time [7, 73, 92]. Micro-interaction EMA (μEMA) has been proposed to reduce this burden by delivering lightweight, single-question prompts on wearable devices [26, 32, 47, 74]. Although μEMA improves feasibility for high-frequency data collection, it does not fully address challenges in sustaining engagement or encouraging rich, detailed responses, particularly in long-term deployments.

2.2 Gamification and data-driven games for engagement

To improve engagement in data collection, researchers have explored incorporating game-like elements into survey interfaces. These approaches typically add lightweight incentives—such as points, leaderboards, or small interactive tasks—to encourage continued participation [38, 53, 86]. While such methods can improve compliance, they augment existing survey workflows rather than fundamentally changing how data collection is experienced, with limited impact on the richness of collected data.

Beyond gamification, other work has explored games that adapt to user data, particularly in the context of biofeedback. These systems modify gameplay based on physiological signals such as arousal or emotional state [15, 22], or introduce new interaction mechanisms based on sensed inputs [54, 62]. While these approaches demonstrate how user data can drive engaging gameplay, they are designed to enhance the gaming experience itself rather than to support data collection. Consequently, prior work either treats games as a thin layer on top of surveys or uses user data to improve gameplay without addressing data collection needs.

2.3 Potential of LLMs for personalized and adaptive data collection systems

Together, these limitations point to an opportunity for new approaches that make data collection itself engaging and sustainable. In particular, systems that leverage intrinsic motivation—rather than external incentives—may enable participants to contribute richer and more detailed data over longer periods.

Recent advances in large language models (LLMs) provide new capabilities for building such systems. Beyond predictive modeling [33, 36], LLMs can perform open-ended sensemaking over heterogeneous data sources, integrating multimodal inputs and generating coherent natural language interpretations [8, 50, 84, 93]. They can also support interactive and user-facing applications, such as conversational agents and personalized journaling systems, enabling continuous, context-aware interactions with users [63, 94, 98]. Unlike traditional pipelines that rely on predefined features or fixed analyses, LLM-based systems can flexibly adapt to new data, refine their interpretations over time, and generate dynamic content tailored to individual users.

These capabilities open the door to personalized and adaptive data collection systems in which user data not only informs analysis but also directly shapes the interaction experience. In this work, we leverage LLMs to generate personalized game content that adapts to user input, making behavioral data collection engaging and motivating richer self-reports.

3 Overview of PAGE

In most data collection studies, participants provide self-reports anywhere from once a week to multiple times a day. While they may receive financial compensation for their efforts, they rarely receive any direct benefits from the data collection process itself. This disconnect can lead to a decrease in compliance and response quality over time. To address this challenge, we introduce PAGE, a novel approach designed to improve participant engagement in self-reported data collection studies by transforming collected data into interactive game elements. The approach attempts to (1) leverage intrinsic motivation in people who regularly play word/puzzle games, (2) enhance gameplay by using data being collected in a study to create tailored in-game elements, and (3) complement the data collection process with additional in-game surveys. In short, PAGE creates a novel, personalized gameplay experience that serves as a feasible channel to collect ***richer self-reports and additional, voluntary data in research studies***.

PAGE has three core components:

- The **game module** is a word or puzzle game that might be enjoyably and regularly played by some study participants.
- The **data collection module** captures self-reports and passive sensing data directly from the participants' devices.
- The **personalization module** reasons about the data collected from the data collection module and uses the data to create tailored game elements for the game module.

To evaluate our approach, we implemented a proof-of-concept system based on PAGE, where the game module used for testing is a variant of a word wheel puzzle game (sometimes also called a word polygon game), using a rule set similar to the Spelling Bee variant published by the New York Times (see Figure 2) [19].

3.1 Game module: A word wheel puzzle game

Word wheel puzzles, sometimes also called word polygon puzzles, target puzzles, or honeycomb word puzzles, present players with a group of letters from which words must be constructed. One ruleset for this game is used in Spelling Bee, a popular variant published by the New York Times (NYT) [19]. NYT games puzzles have a

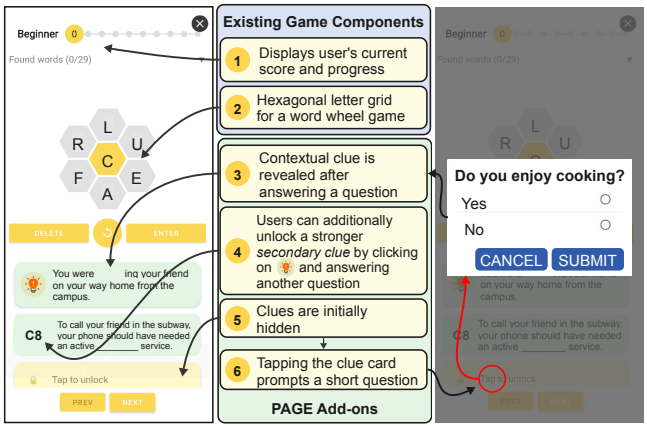


Figure 2: User interface of our adapted word wheel puzzle. PAGE allows: (1) personalization of game elements (e.g., clue cards) using collected data; and (2) built-in mini-survey to collect additional data during gameplay (e.g., to unlock a clue, the user must answer a quick question).

massive user base, with over 10 million daily players and more than 11 billion total plays in 2024 [72, 82]. Although PAGE could be applied to any popular word game format, such as crossword puzzles, anagram puzzles (e.g., Jumble), word wheel puzzles (e.g., Spelling Bee, Target), or word deduction puzzles (e.g., Wordle), to test the feasibility of the idea, we implemented a modified version of the famous puzzle game Spelling Bee.

In word wheel puzzles, players must guess valid English words that are at least four letters long, using only a provided set of seven letters, one of which must be the highlighted center letter. Although the version published by the NYT lacks built-in hints, many players voluntarily author and share custom clues for difficult words on help forums [20]. PAGE uses an LLM-powered backend to generate personalized clues using participants' sensing data from the day before, replacing a generic hint with an engaging, personalized reflection.

To minimize the learning curve in our study, we created an interface that is close to the NYT version (Figure 2). The upper half of the screen contains the hexagonal letter grid as in the online NYT implementation, but the lower half of the screen in our implementation contains a list of clue cards players can use to unlock hints to help them guess the words that can be formed. The search space of possible letter combinations is enormous, and players can get "anchored" to subsets of the solution space; clues can help players move beyond the words they have already found, which sometimes block other ways of thinking. The clues our system provides are initially hidden, and participants can tap the cards to unlock the contextual clues. There are two levels of clues: (1) the contextual clue is a "fill-in-the-blank" hint connecting the target word to the user's activities from the previous day (e.g., For CALL:"You were _____ing your friend at 10 AM"); and (2) a second, more revealing clue that participants can additionally unlock, giving the starting letter and total character count of the target word (e.g., "A5" for the word APPLE).

Table 1: Data streams and their sampling frequencies

Device	Data Stream (Sampling Frequency)
Phone	GPS location (1m), app use (1m), lock/unlock events (1m)
Smartwatch	Step counts (1m), ambient noise classification [24] (5m), heart rate (1m), μEMA responses (15m), skin temperature (10s), wrist movement data (10s).

3.2 Data collection module: μEMA and wearables sensing data

To collect self-reports, we used a multimodal μEMA (micro-EMA) system [48, 49]. Multimodal μEMA allows users to answer smartwatch prompts using either touch or voice inputs (Figure 4). We chose this approach to capture rich, spoken details from participants while still giving them the flexibility of quick, silent screen taps when they cannot speak out loud. The smartwatch delivered vibration prompts every 15 minutes from 8 AM to 10 PM daily. When participants felt the vibration, they reported their current posture and physical activity. They had 10 s to respond to each prompt and could either speak their answer (e.g., *"I am sitting and working on my laptop."*) or tap one of the single-word options on the screen (e.g., *Sitting, Conversing, Walking, Using Phone*). Details about the implementation of the touch-based interaction are included in Appendix A. We also collected passive sensing data from both the smartphone and the smartwatch to complement these active self-reports (see Table 1).

PAGE also allows researchers to collect additional data through in-game surveys. In our PoC, when a participant taps a clue card to see a hint, the app first shows a short question (see Figure 2). Answering the question unlocks the clue, so participants exchange responses for progress in the game. We populated these in-game surveys with questions such as: what they were doing when they were not wearing the watch or when they did not respond to μEMA prompts, label places like home or work, or answer simple questions about their habits. This approach is flexible and can be adapted to include any survey or questionnaire researchers need. For the full list of clue unlock questions we used, see Appendix B.

3.3 Personalization module: Using an LLM to generate personalized in-game elements

In our game generation module, we generate the daily puzzle by (1) synthesizing a narrative diary from the previous day's data, (2) generating a word wheel puzzle vocabulary list, and (3) asking an LLM to create a personalized contextual clue for each word using the diary as context (Figure 3).

First, the system aggregates multimodal μEMA and passive-sensing data collected from wearable and mobile devices. The data are chronologically sequenced and provided as context to the LLM, which synthesizes a cohesive, narrative-style diary reflecting the user's previous day (Figure 3). For the full, detailed prompts used to generate the narrative-style diary, see Appendix C.

The module uses the generated narrative-style diary to personalize in-game elements. In the PoC, PAGE generates the daily

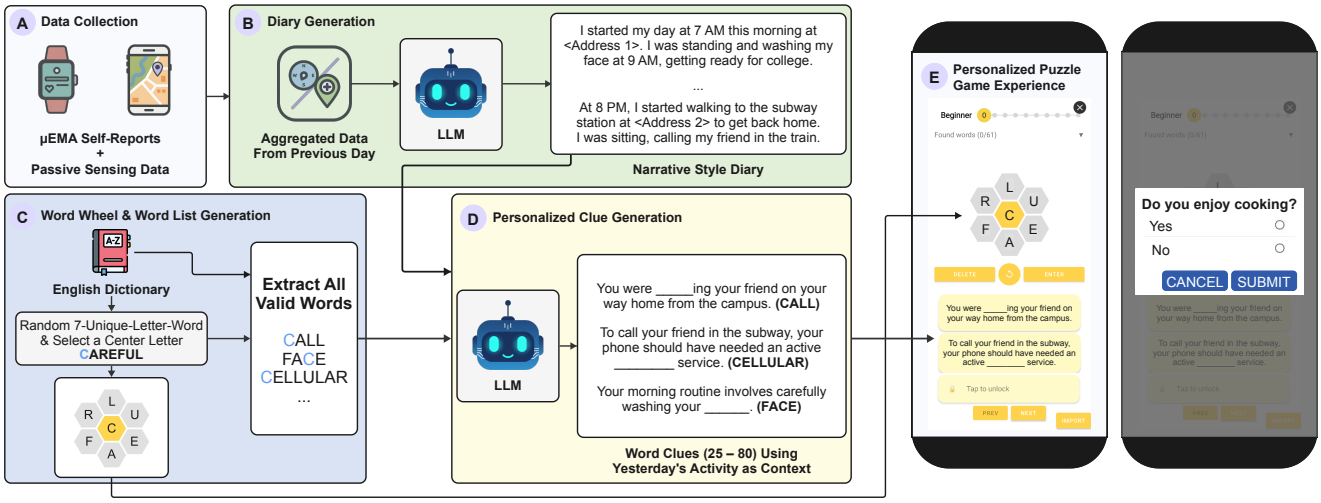


Figure 3: System Overview

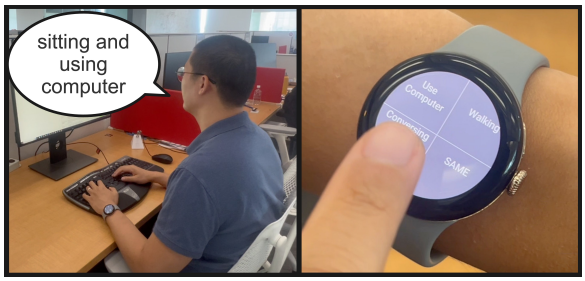


Figure 4: Multimodal μEMA allows participants to self-report via: (1) tapping on an option on the screen, or (2) providing more detailed responses via speech interaction.

word wheel vocabulary list (Figure 3C). The system first queries an English dictionary to randomly sample a seed word with exactly seven unique letters (e.g., "CAREFUL") and randomly selects one as the mandatory center letter. The system then extracts all valid dictionary words that are at least four letters long, use only those seven letters, and contain the center letter. The system also excludes inflected word forms (e.g., words ending in "-ed" and "-ing") from the target vocabulary. To ensure playability, the system removes all obsolete terms (words with "[Obs.]" tag in the dictionary) and words with low frequencies using the Python wordfreq library [79]. Finally, we prompt the LLM to remove all profanities, human names, proper nouns, and obscure jargon (e.g., specific subspecies of an animal or rare ethnic instruments) that are inappropriate for a general puzzle audience. For more details, see Appendix D.

Finally, the synthesized narrative diary and the filtered vocabulary list are provided to the LLM to generate a personalized contextual clue for each target word (Figure 3D). These clues are formatted as fill-in-the-blank style sentences that semantically anchor the target word to the user's activities and experiences from the previous day. For instance, if the LLM-generated diary contains calling a friend on the phone during the day, and the puzzle contains the word "PHONE," the LLM might generate the following contextual clue: *"You were likely using your _____ to call your friend last night."* For more details on how we generate the clues, see Appendix E.

3.4 Implementation

We implemented our personalized word wheel puzzle game in a custom Android application (Android 9+). After all data from the previous day had been uploaded to our research server, the personalization module ran on that same server each day at 3 AM local time to generate personalized puzzles. The generated puzzles were formatted as JSON files and uploaded to our Google Firebase database. The Android app continuously checked for updates and automatically downloaded these files to the local device as soon as they were ready. The downloaded daily puzzle remained available until midnight, and the app saved all progress so participants could play across multiple sittings throughout the day.

We implemented the multimodal μEMA application for Android WearOS smartwatches. We transcribed the μEMA audio recordings using Google Cloud Speech-to-text API [23], and ran Google YAMnet model locally on the watch for ambient noise classification [24].

For the personalization module, we used Google DeepMind's Gemma3 (27b) large language model for all generative tasks to process data locally and protect participant privacy [34].

4 Feasibility user study

We conducted a feasibility user study with 15 participants over two weeks. The goal of the study was to (1) explore the feasibility of our system for real-world deployment, (2) assess the effects of adaptive games on engagement and data quality/quantity, and (3) collect participant feedback on the acceptance and usability of PAGE to improve it for future deployments.

4.1 Recruitment and participant compensation

Our study protocol was approved by the Institutional Review Board at Northeastern University (IRB#24-02-47). Participants eligible for

Table 2: Participant demographics

	N = 15
Age (years)	M = 26.5 (SD = 3.3)
Sex	Male 9 (60%), Female 6 (40%)
Race / Ethnicity	Asian 12 (80%), Black 1 (7%), White 1 (7%), Other 1 (7%)
Frequency of playing Word Puzzles	Very Often 5 (33%), Often 4 (27%), Sometimes 3 (20%), Rarely 3 (20%)
Familiarity with Spelling Bee	Extremely 2 (13%), Very 4 (27%), Moderately 2 (13%), Slightly 3 (20%), Not at all 4 (27%)

our study were (1) aged 18 or older, (2) fluent in written and spoken English, (3) used Android smartphones, (4) were willing to wear a smartwatch for up to 14 days, (5) did not have any cognitive or hearing impairments, (6) were able to read the phone screen without reading glasses, and (7) self-reported regularly playing and enjoying word puzzle games. Since PAGE builds on games participants are familiar with and adapts them into a personalized experience, we recruited participants who said that they "play word puzzle games (for example, Wordle, Mini Crosswords, or Spelling Bee)" at least sometimes, i.e., *sometimes (1–3 times per month), often (1–3 times per week), or very often (daily or almost daily).* Given the feasibility nature of the study, recruiting those unfamiliar with such games would be less effective than testing the system with users who already have a baseline interest in the genre. We discuss the implications of this targeted recruitment in Section 7.

Participants were recruited using university mailing lists, social media announcements, and physical flyers posted in on/off-campus locations. Participants were compensated up to $54 ($20 each for pre-study/post-study interviews and $1 for each study day that participants adhered to the protocol). Compensation was distributed electronically via Amazon gift cards at the conclusion of the study.

Our study employed a within-subjects design with two conditions, each lasting one week. In the *puzzle condition*, participants were asked to wear the provided smartwatch during their waking day, report their in-the-moment posture and physical activity labels when prompted, and optionally play the daily word puzzles delivered to their smartphones. Protocols for the *control condition* were identical to the puzzle condition, except that participants did not receive daily puzzles during that week. We randomized the order of each condition: participants were assigned to the *puzzle first* group (N=8) or *puzzle later* group (N=7) using block randomization. Participants in the *puzzle first* group received the daily puzzles during their first week of data collection, whereas those in the *puzzle later* group had their daily puzzles delivered in their second week.

4.2 Study protocol

The study consisted of three phases: an in-person pre-study session, a data collection period of 14 days, and an exit interview. During the pre-study session, our research staff obtained signed consent, collected responses for the demographic survey, and installed the Android app on the participants' smartphones.

During the data collection period, we provided participants with an Android Wear smartwatch to wear during their waking day for 14 days. Between 8 AM and 10 PM each day, the smartwatch delivered a vibration prompt once every 15 min. We showed participants a demo video on how to respond to the μEMA prompts and how to play the game (demo video included in supplemental materials). We encouraged participants to verbally report their responses whenever possible, including as many additional contextual details as possible (e.g., where the activity is occurring), because this would provide richer, more informative posture and activity labels than the static labels displayed on the screen. We also invited participants to complete the optional daily burden survey in our application in the evenings, whenever possible, to gain day-level insights into participant burden and puzzle enjoyment.

During their puzzle week, participants were notified via an in-app notification when a new puzzle was ready to play. The daily puzzles were available for the entire day until they reset at midnight, and participants were able to revisit the puzzles multiple times throughout the day because their progress was automatically saved.

Upon completing data collection, we invited participants to an exit interview session that lasted one hour. We measured usability of our system with the standard SUS and NASA-TLX surveys. We conducted semi-structured interviews with participants to get participants' qualitative feedback on their experience in the study and suggestions on how to improve the system in future deployments.

5 Results

In this section, we report the quantitative and qualitative findings from our user deployment.

5.1 Supporting participants' μEMA engagement with context-aware puzzle games (RQ1)

We investigated whether PAGE supported EMA engagement and improved data quality.

5.1.1 Impact on participants' EMA response rate. During the user study, we collected 210 days of data from 15 participants. Participants responded to 5,286 of the 6,821 delivered prompts, with 3,706 (70%) via speech interactions. During the puzzle week, they answered 2,362 of 3,016 prompts (1,650 via speech; 69.9%). During the non-puzzle week, they answered 2,924 of 3,805 prompts (2,056 via speech; 70.3%). We compared the participants' compliance with the μEMA prompts between the two weeks using two commonly used prompt response metrics: compliance rate and response rate (see Table 3) [47, 49]. Response rate reflects participants' engagement with delivered prompts, while compliance rate captures how much data were collected relative to what was scheduled:

$$\textit{Compliance rate} = \frac{A}{S} \times 100\%; \quad \textit{Response rate} = \frac{A}{D} \times 100\%$$

where A is the number of answered prompts, S is the number of scheduled prompts (8 AM–10 PM), and D is the number of successfully delivered prompts.

While the overall response and compliance rate is slightly lower for puzzle week, results from the paired t-test show no statistical differences between the conditions ($p > .05$). The main reason for the discrepancy between the compliance rate and the response rate is the smartwatch battery life. The μEMA prompts were scheduled from 8 AM to 10 PM (14 h/day), but during the study, the battery life

Table 3: Prompt response metrics between puzzle week and non-puzzle week (between participants).

Metric	*Puzzle week*		*Non-puzzle week*	
	Mean	SD	Mean	SD
Compliance rate (%)	48.0	14.4	53.3	15.1
Response rate (%)	75.9	13.5	77.1	11.0

lasted for an average of 10.37 hours (SD = 2.38) during the puzzle week, and 11.01 hours (SD = 2.14) during the non-puzzle week. The limited battery life of the smartwatch can be attributed to either a hardware issue or, in some cases, participants forgetting to charge the watch the day before.

We further examined whether response rate changed across the days within each study week and if that change differed between the puzzle and no-puzzle conditions (see Figure 5). We used a linear mixed-effect model with random intercept and slope for each participant.

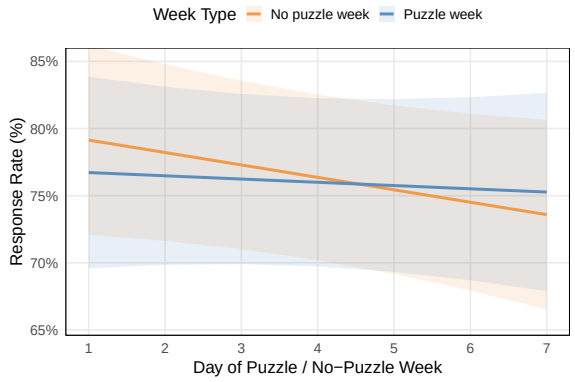


Figure 5: Daily response rate trends. Response rates gradually declined over time, but the drop was slightly faster during the non-puzzle week than the puzzle week.

While we visually observed a steeper decline in response rate during the non-puzzle week, it was not statistically significant. Average response rates declined by 0.93% per day during the no-puzzle week and 0.24% per day during the puzzle week.

5.1.2 Impact on participants' perceived EMA burden. We collected 101 responses from the optional daily burden survey (response rate: 48%). Among the responses, 37% mentioned the participant had to charge the watch in the middle of the day to keep it running. We show the distribution of responses for each question in the burden survey for both conditions in Figure 6. A χ^2 test showed no statistical significance in reported burden between the two weeks ($p > .05$).

5.1.3 Impact on participants' EMA response behaviors. To quantify the richness of participant label responses, we used a process called semantic parsing [60]. Because traditional NLP metrics such as word counts cannot reliably capture differences in *information richness*, we examined the amount of information within a spoken μEMA response. We define a semantic tag as one distinct piece of contextual information about the participant. For example, a response "*Sitting in my room, eating an apple*" could be broken down into the following tags: *Sitting* `[posture]`, *my room* `[location]`, *eating* `[activity]`, and *apple* `[object]`. We list different tag categories and unique tag counts in (Table 4). We also provide a more detailed explanation of the parsing process in Appendix G.

Table 4: Comparison of unique semantic tags between puzzle week and non-puzzle week.

Tag	*Example*	*Puzzle week*	*Non-puzzle week*
Posture	*Sitting*	15	16
Activity	*Talking*	285	267
Objects	*Laptop*	119	109
Location	*My room*	198	179
Time	*At 7 PM*	24	14
Social	*With friends*	28	14
Purpose	*To do homework*	57	51
State	*Hungry*	2	6
Specificity	*Watch "Friends"*	33	33
Total		**761**	**689**

Overall, we collected 72 more unique contextual tags in the puzzle week compared to the non-puzzle week. Participants reported more unique tags for activity, objects, location, time, social contexts, and purpose in the puzzle week.

During the semi-structured interviews, all participants claimed to notice the personalized clues. While personalization did not change EMA compliance, seven of 15 participants (47%) found it motivating in some way. Specifically, five explicitly said they made extra efforts to provide more detailed answers to make the clues more interesting: *"I started adding more specific details on where I am, what I'm doing, because I was expecting—how will the AI integrate all these things into my personal clues tomorrow?" [P08].* A sixth participant noted that personalization made them *"more aware and conscious of" [P04]* their daily activities to prepare for the puzzle. A seventh participant said they *"[played the personalized game] more often than the original Spelling Bee... because it's kind of more personalized to [them]" [P15].* The remaining participants did not explicitly change their response behaviors, though they too noticed the personalization. For P03, the puzzle and the prompting remained two separate activities: *"No, actually. For me, the puzzle and the prompt, these were basically separated." [P03].* These findings suggest that personalized puzzles can motivate richer and more detailed self-reporting, but the effect is not universal. Also, because we recruited participants who already enjoy word puzzles, we cannot fully separate the effect of personalization from participants' general enjoyment of the genre.

5.2 Participants' interactions and engagement with PAGE (RQ2)

Across participants, the median number of days the game was played was five out of seven (71%; IQR = 2.5). On average, participants spent 22 min/day playing the game (SD = 22.77). On 40 of 74 days (54%), participants completed gameplay in a single session,

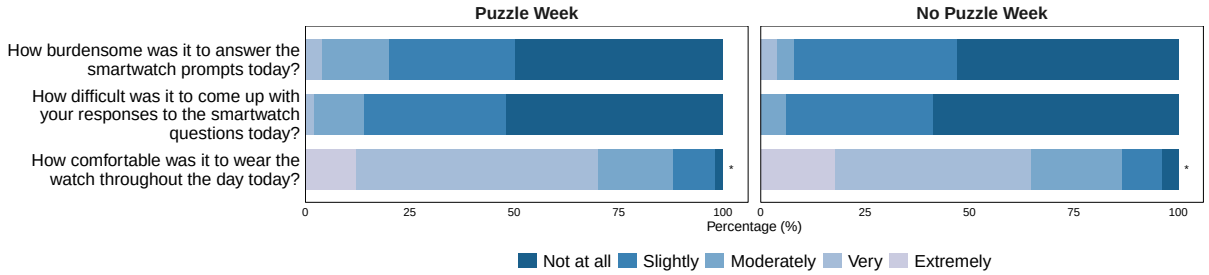


Figure 6: Distribution of responses for individual items related to smartwatch prompts in the daily burden survey. Survey items with an asterisk (*) are reverse coded.

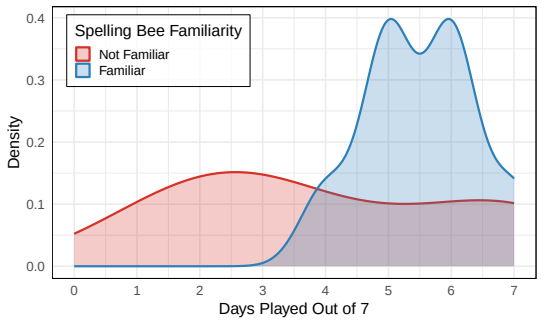


Figure 7: Participants who were already familiar with Spelling Bee prior to joining the study tended to play our adaptive game more often.

whereas on the remaining 34 days (46%), participants played the game multiple times throughout the day. Overall, we found that participants who were familiar with, and had played, the New York Times Spelling Bee game regularly before joining our study tended to play our adaptive game more frequently than those who were not familiar (Figure 7). PAGE received a high mean usability score (SUS) of 82.5 (SD = 11.26) and low NASA-TLX score of 38.20 (SD = 14.45), suggesting high perceived usability and low effort. We present the results from the individual questions of SUS and NASA-TLX, as well as results from the related items in the daily burden survey in Appendix F.

Participants correctly guessed an average of 20.6 words per puzzle (SD = 15.3), and unlocked an average of 16.97 clues per day (SD = 17.8). Leveraging the *clue unlock* functionality, **we were able to collect an additional 1,188 in-game survey responses**. This corresponds to an average of 79 surveys per participant, provided voluntarily while playing the game, *in addition to the prompted μEMAs*, without expectation of financial compensation. Because these responses are voluntary, embedded in gameplay, and surfaced only when a participant chooses to unlock a clue, this in-game channel is not directly comparable to fixed-interval smartwatch μEMA on standard metrics such as compliance; it is therefore better read as a feasibility signal than as a between-condition gain in compliance rate. Still, these additional responses suggest that in-game surveys could offer a viable channel for collecting additional data that warrants further study over longer deployments. Figure 8 shows an example of how these additional in-game questions helped supplement the missing μEMA self-reports.

Overall, participants expressed positive opinions on the word puzzle, with one noting they were *"excited to play the game every day," [P08]*; others found the personalized clues relatable because they contained *"more details about [their] specifics and felt more personalized to what [they were] doing." [P01]*. Twelve out of 15 participants (80%) preferred the puzzle week more than the no-puzzle week. Also, nine out of 10 participants reported feeling little-to-no burden when playing the game every day, and one participant (P13) reported a minor burden only because they owned two smartphones (iPhone/Android) and our app was installed in their secondary Android device, which they had to constantly carry and keep charged: *"I felt like it was just an additional task. It wasn't burdensome at all." [P10]*. Participants were particularly interested in how the games incorporated their self-reports into the in-game elements: *"The other day I was reading a Nature article and then there was a clue about me as a bioinformatician reading a Nature article… I was surprised, like, it took all that into consideration." [P08]*.

Participants' opinions on having to answer short questions to unlock the clues were split. Eight out of 15 participants reported feeling no burden: *"No, I don't think so. I was like, okay, this is something I have to trade, so that's fine… The process of unlocking the clues, I looked at them like they are trying to help me guess the words and not the other way around." [P02]*. Seven out of 15 participants responded feeling disrupted having to answer a question in the middle of the game: *"I guess to some extent, because the additional pop-up questions, they do kind of seem a bit like irritating … you have to go through and search for what it actually was, and also specifically for the cases when you don't really remember—that breaks the flow even more." [P03]*

Participants' engagement with the game was shaped by a recurring tension between motivation and difficulty. Eight out of 15 participants played opportunistically, filling idle moments, e.g., on public transit, during lunch breaks, or while waiting, rather than scheduling dedicated sessions: *"Usually when I travel, I'm on the public transport... mindlessly scrolling. So that is when I was like... let's play this game instead of [scrolling] through reels." [P02]*.

Engagement decreased when difficulty spiked, and participants described three distinct sources of frustration. The first was technical: system limitations occasionally caused the puzzle to reject

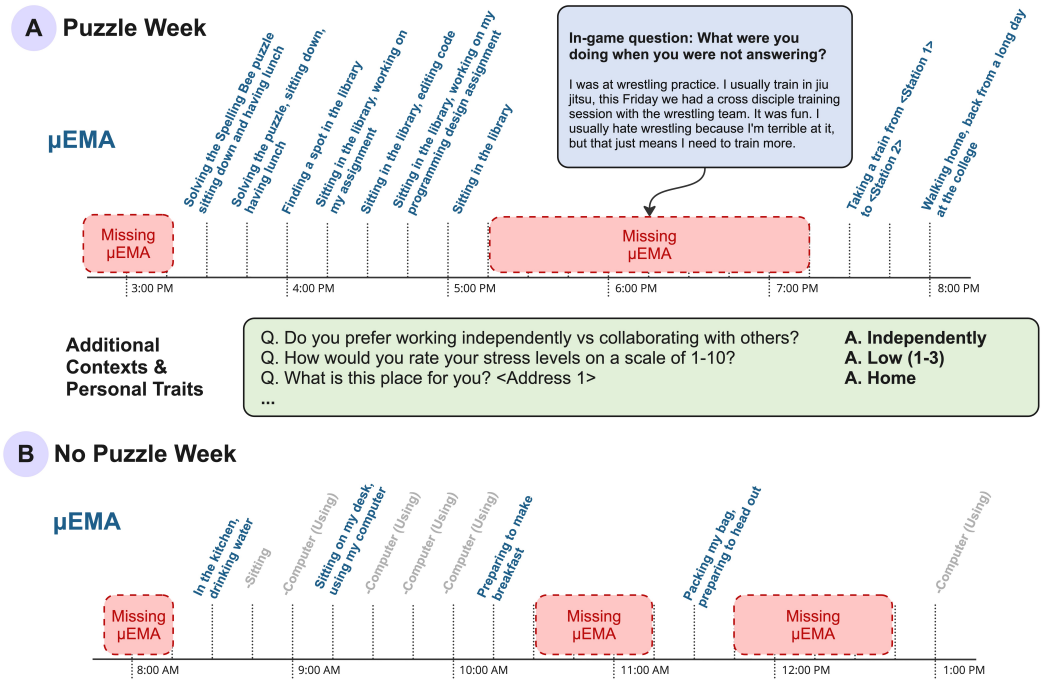


Figure 8: The visualization of data collected during puzzle and no puzzle weeks from one of the participants. The participant provided richer self-reports during the puzzle week compared to non-puzzle week, and answered additional in-game questions.

valid words, or produced clues that felt incoherent and disconnected from participants' actual experiences. *"There were certain instances where the word exists, but the game would not accept it as an answer. That was an issue." [P02].* Some LLM-generated clues felt forced rather than meaningful: *"The [clues] themselves, many times, [...] becomes hard to understand or what it is actually referring to." [P03].* The second source of frustration was the number of words per puzzle. When the total word count was too high, participants reported losing motivation before they had even begun, because completing the puzzle felt unachievable from the outset: *"Once I know there were 76 words, I just practically lose interest that day because I know I'm not going to finish [sic]." [P08].* This frustration was compounded by the psychological weight of near-completion: *"I could reach only to 96 or 98. That was hard because if you played regularly, you wanted to do it completely and do not lift it midway. So that makes you feel incomplete [sic]." [P04].* The third source was general word-guessing difficulty, particularly with longer words. Participants described a consistent threshold beyond which the game stopped feeling rewarding and became a reason to disengage entirely, with consequences that extended to the following day: *"If it's very difficult where I cannot guess what that is even going to be, that would discourage me from continuing to spend time with that, and next day I'm more likely to not go to it at all." [P01].* Some participants described self-imposing a stopping point once they felt stuck rather than pushing through: *"If I find it really difficult, I just like chuck it, let it go. I won't play anymore." [P01],* while others noted that difficulty needed to be calibrated in both directions: *"If it's too hard, then you'll naturally lose your interest in playing... even if it's too easy, it's just not a really interesting thing to do." [P08].* Conversely, when difficulty was well-calibrated, participants had a sense of accomplishment that sustained engagement: *"I was very happy [and] proud." [P01].*

5.3 Participants' attitude and acceptance towards context-aware puzzle games (RQ3)

Participant interviews suggest designers should consider privacy implications, gameplay experience, and compliance behaviors in self-reporting.

5.3.1 Privacy concerns. Participants held notably different attitudes toward privacy depending on whether in-game clues were derived from their active self-reports or from passively collected location data. Broadly, participants were more comfortable with clues generated from their own spoken reports, because active reporting carried a sense of conscious ownership and informed consent over what was being shared: *"I volunteered to give [that] information, so that's fine." [P02].* As P01 explained, *"I was pretty comfortable because ... I had the knowledge of what [the data was] going to be used for. That made the whole process smoother for me." [P01].*

Location-based clues, however, triggered stronger discomfort, particularly when they referred to places participants had merely passed through without consciously registering. P02 described a reaction to the prospect of passive location data appearing in the game: *"If a game gave me a clue, like, you just walked at [Student Center]—that would be really horrifying for me." [P02].* Prior consent appeared to substantially mediate this discomfort. P01 drew a direct contrast between knowing and not knowing that location would be tracked: *"It did not make me feel uncomfortable because [the*

researchers explained location tracking] beforehand. If I did not know beforehand that [the system was] going to track my location, and then [the in-game question asked]—what is [anonymized location] to you? [It would be different]. But because I knew the context, it was not that uncomfortable for me." [P01].

Participant comfort with passive sensing is not simply a matter of initial disclosure, but reflects a broader need for ongoing control over what data is collected and how data are used in the game. Rather than relying solely on upfront consent, systems of this kind may benefit from giving participants granular agency over their data—for instance, allowing them to opt out of location-based clue generation in favor of self-report data, or to review and approve what contextual information is used before it appears in gameplay.

5.3.2 Engagement and compliance behaviors. Both sensing modalities produced engagement through different mechanisms. Location data generated recognition-based engagement—participants were surprised to see familiar places and routines surface in the puzzle: *"The game really captured what I was doing—the places that I was, the locations, as well as the shows that I was watching. So that was kind of exciting." [P10].* Self-report data, by contrast, produced a more reflective form of engagement: *"[During] the first week I was... slacking and not doing anything... [and my μEMA responses were] mostly lying on the bed and using my phone... Then I realized I should work out more." [P16].* For some, this reflection dynamic also fed back into reporting behavior, motivating richer and more deliberate self-reports during the puzzle week: *"I started adding more specific details on where I am, what I'm doing, because I was expecting—how will the AI integrate all these things into my personal clues tomorrow?" [P08].* This interplay between self-reporting and reflection demonstrates how the act of self-disclosure can facilitate a reflective experience and sometimes motivate richer data entry in anticipation of more tailored game content—creating a feedback loop that passive sensing alone cannot trigger.

5.3.3 Clue difficulty and interpretability. The two sensing modalities diverged most sharply on clue interpretability. Clues derived from active self-reports were generally easier to connect to the target word, because participants could directly trace a link between something they had consciously described and the clue that followed: *"The prompts or the clues that come from what I was actually doing over the day, it kind of seemed a bit easier to grasp." [P03].* Location-based clues, by contrast, were more frequently experienced as opaque, because GPS data often captured places participants had passed through incidentally rather than visited intentionally: *"When you were actually moving through the day, many a times you don't even really notice—okay, I'm actually near this place. So that kind of makes it harder to kind of decrypt." [P03].* At its most extreme, some participants perceived location-based clues as incoherent rather than personalized: *"The system is forcing the word onto an experience that I had... it doesn't have to be that way." [P08].* These findings do not suggest removing location data from the pipeline, but rather that the system needs greater contextual intelligence about which passive data points are meaningful to the participant. A longitudinal deployment could help address this; over time, the system could learn to distinguish between places participants intentionally frequent and those they merely pass through, reserving location-based clues for the former.

5.3.4 Interest in continued use. We surveyed participants' perceived long-term value of PAGE in exit interviews. Fourteen of 15 participants (93%) expressed interest in continuing to use the app outside the study. Some conditioned this on better schedule adaptation (e.g., *"I would if the frequency of the prompt is less and it's more adapted to my daily schedule" [P08]*) or enhanced data control: *"I would be more comfortable if [PAGE gave] me more control over... what kind of data I'm sharing" [P06].* Some were motivated by enjoyment, stating they would be *"pretty happy to do so because the whole experience was fun" [P13].* P06 anticipated that the puzzles would become more engaging over a longer deployment, imagining a system that would keep learning about them as they played: *"It felt like the puzzle was learning about me as I play more of it or as I... participate in the study more. So I did feel like... instead of one week, if maybe it was over like a month or two months, [the puzzles] would get more and more fun to play with. Maybe it might feel like a friend or a companion."*

Taken together, participants were generally receptive to using PAGE beyond one week, although evaluating the system's actual effects on longitudinal engagement warrants further study.

6 Discussion and Future Work

We discuss the implications of our work for longitudinal studies, design guidelines for future personalized adaptive games, and expanding PAGE to additional game types for a wider audience.

6.1 Using personalization to support engagement and voluntary data

Prior EMA research highlights the burden of repeated surveys and a reliance on financial incentives to sustain engagement [4]. While gamification has been explored as a method to improve compliance, it is often limited to attaching surface-level rewards to compliance, with limited impact on data quality [38, 53, 86].

Our approach instead leverages intrinsic motivation by using collected data to personalize a game that participants already know. By using participants' self-reports to personalize the game, many participants found the experience more interesting and personally meaningful, and some contributed more detailed responses as a result. In addition, we introduced in-game surveys that allow participants to unlock hints or progress by answering brief prompts, creating a secondary, voluntary channel for data collection. This yielded nearly 1,200 additional, uncompensated responses.

Together, these mechanisms improve data richness and coverage. Their effectiveness, however, depends on how salient the connection between a participant's own experience and the resulting clue is. Participants generally found clues drawn from their own self-reports clearly connected to their experience, whereas clues drawn from passively sensed data were often harder to recognize as personally relevant. Future designs should therefore strengthen this connection by prioritizing data types that participants both understand and feel ownership over.

6.2 Toward adaptive longitudinal data collection

Traditional EMA systems are largely static, relying on predefined, close-ended surveys that are repeatedly administered over time [51]. While effective for measuring specific constructs, this approach

limits the richness of collected data and does not adapt to changes in user behavior or context.

In contrast, our findings show that integrating data collection into a personalized game can motivate participants to provide richer and more contextualized information. Rather than a generic game experience, user data dynamically shape the gameplay, which could allow the system to continuously accumulate more detailed data and better reflect users' evolving routines and preferences over time. Crucially, because the success of many longitudinal interventions relies on self-reports to track progress, approaches like PAGE are relevant for future behavioral and health applications.

Looking forward, PAGE opens opportunities for more adaptive and autonomous data collection. With recent advances in LLM-based sensemaking [8, 45, 46, 52, 95], future iterations of PAGE could continuously analyze incoming data, learn about individual users, and dynamically adjust game difficulty or themes based on inferred user preferences, while also generating targeted μEMA prompts or in-game surveys to fill in gaps or clarify uncertain behaviors [9]. This closed-loop system, where data informs personalization, and personalization in turn drives further data collection, could enable more efficient, user-centered, and scalable longitudinal studies. Furthermore, PAGE's mechanics can be adapted to non-puzzle genres. With game developers turning their attention toward exploring LLM-generated, personalized storylines [39, 64, 66], future work could integrate PAGE into these narrative games and collect important longitudinal data across a wider audience.

6.3 Future Design Considerations

6.3.1 Challenges for longitudinal viability. Future work should address the limitations of our system identified in the initial feasibility study. A central open question is how to sustain engagement once the initial novelty wears off. One direction is to offer participants varied content and choice [30, 40], which, for our system, could include multiple puzzle variants across different genres.

This decline in novelty stems in part from clue monotony and a learning effect, as participants become accustomed to the mechanics over time. Future systems may consider adjusting difficulty by providing stronger assists (e.g., revealing partial words instead of only the first letter) for novices, while offering more challenging clues for experienced players to maintain interest [61, 70].

Because nearly half of participants also found the in-game clue unlock questions disruptive to their gameplay, future designs must carefully calibrate the cadence, content, and timing of prompts based on user preferences [42, 81]. Alternatives could include lowering the prompt frequency, strictly using multiple-choice formats to minimize cognitive load, or adding an option to answer questions before launching the game to preserve the flow of play [46].

6.3.2 Data privacy and informed consent. Our qualitative findings demonstrate how personalized games can act as a tool to help participants better understand their own data footprint. While most participants demonstrated positive ownership over self-report-based clues, passive sensing-based clues sometimes heightened their awareness of the potential invasiveness of passively sensed data [43, 57, 88]. Future research might explore how personalized games can communicate to participants, in concrete and layperson-friendly ways, exactly what can be inferred from each data type (e.g., the clues it generates). By coupling this transparency with the ability to dynamically opt in and out of data gathering, future systems could transform this privacy friction into an opportunity for deeply informed longitudinal consent, which may be especially crucial in studies collecting sensitive data such as health records [87].

6.3.3 Ethical implications. Results from our user study show the feasibility of leveraging AI-driven personalization to improve engagement [80]. Such personalization, however, carries a dual-use risk: the same tailoring meant to support participants could be exploited to maximize engagement against their interests (e.g., to extend ad exposure or promote endorsed content). Future systems must remain strictly user-serving by being transparent about which data are used for personalization and the associated risks, and giving users granular control over what types of data are reflected back to them. Finally, system success must be measured not solely by researcher-centric metrics such as engagement time or data volume, but also by user-centric metrics such as perceived enjoyment, utility, and intrinsic motivation to continue engaging [28, 37, 71, 90].

7 Limitations

This early-stage feasibility work has several limitations. First, we tested only one word wheel puzzle variant with a small group of participants over a short period (one week per condition), so we could not extract statistically significant results on response rate or evaluate longitudinal effects. Moreover, because we recruited participants who already enjoy word puzzles, we cannot fully attribute our findings on preference and engagement to personalization rather than to enjoyment of the puzzle itself.

The Pixel Watch we used also suffered from poor audio quality and limited battery life. Newer smartwatch models may alleviate these hardware constraints. Finally, we used a small, open-source LLM for participant privacy and sustainability. This model, however, occasionally generated clues that were too wordy or obscure, which some participants noted made the game harder.

8 Conclusion

We introduced PAGE, a novel LLM-powered approach that transforms routine data collection into personalized games. Our feasibility study demonstrated that tailoring gameplay with participants' own passive and self-report data is a viable approach that may foster voluntary, curiosity-driven engagement. Overall, PAGE shows promise for supplementing data collection with a rewarding and enjoyable participant experience.

Acknowledgments

This research was partially supported by the Khoury College of Computer Sciences, Northeastern University. We also thank our participants, internal testers, anonymous reviewers, and the members of the mHealth Research Group and UbiWell Lab for sharing their valuable feedback on this project.

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A Appendix A: Activity Level Estimation

To decide which activities to display on the screen, we use a rule-based heuristic algorithm. We determined the participant's activity level using the heart rate (HR) sampled immediately before the prompt: for *Sedentary* activity level, the watch screen displayed the following options: *Computer (Using), Phone (Using), Conversing, and Sitting*. For *Moderate* and *Vigorous* activity levels, the watch screen displayed the options: *Walking, Running, Cycling, and Lifting Weights*. The four options displayed in each scenario and the threshold were determined based on prior work (see Table 5) [49].

Table 5: Heart rate (HR) cutoffs for physical activity levels.

Activity Level	*Heart rate (HR) Thresholds*
Vigorous	HR $\geq$ 140
Moderate	100 $\leq$ HR <140
Sedentary	HR < 100

B Appendix B: Clue Unlock Questions

The in-game clue unlock questions serve as the mechanism for our secondary data collection loop. Whenever a participant attempts to unlock a Tier 1 or Tier 2 clue, the application displays a pop-up micro-survey prompt. These questions are designed to provide researchers an additional opportunity for data collection by treating survey responses as a transactional currency to unlock additional hints. These clue unlock questions fall into five categories:

- **Cluster Labeling:** Displays stationary clusters identified from the location data and asks users to provide a semantic label (e.g., home, work, school). Two consistent responses lock in a finalized label, which is then fed back into the LLM pipeline for future puzzles.
- **Data Gap:** Identifies time periods between 8 AM and 10 PM where the participant either was not wearing the smartwatch or missed more than two consecutive μEMA prompts. Users are prompted to actively recall their activities during those data gaps by actively typing or speaking their response.
- **Priming Clue Questions:** Probes participants' personal traits or habits (e.g., "Are you a morning person?") to dynamically seed the priming clues for the subsequent day's puzzle.
- **Validated Clinical Surveys:** Acts as a placeholder for any *validated survey* that could be inserted for standardized longitudinal assessments. In this prototype, we included items from the PHQ-9 survey [41].
- **UX Surveys:** Probes users' perceived difficulty of the puzzle and their overall satisfaction with the puzzle interface.

Each specific question is displayed only once per day. Furthermore, the presentation order is randomized so that participants never receive the same category of clue unlock questions twice in a row.

C Appendix C: The Narrative Diary Generation Pipeline

PAGE's **personalization module** aggregates the μEMA and passive sensing data collected from the **data collection module** and uses an LLM to generate a narrative-style diary of the user's previous day.

The module first structures raw multimodal data streams into discrete chronological format optimized for comprehension. To achieve this, we first used the DBSCAN algorithm [18] to segment the users' GPS logs into stationary clusters and transitional periods. Subsequently, we used OpenStreetMap's Nominatim API[1] to label each cluster centroid with an estimated building type (e.g., residential, commercial) [69]. Using these spatial blocks as a chronological backbone, we integrated the remaining data as additional environmental contexts. We further queried the OpenStreetMap API to identify nearby Points of Interest (POIs) - such as restaurants, cafes, libraries, theaters, or landmarks which the users *may have noticed during their stay/transition*. Continuous numerical data streams such as step counts, heart rate, and accelerometer samples were translated into discrete contextual tags (e.g., *High Heart Rate, Extended Phone Use, Walked and Rode Vehicle*) to complement the location context and μEMA self-reports.

Finally, this synthesized timeline was formatted into discrete XML blocks. We explicitly formatted the prompt context for narrative diary generation using XML blocks, because it provides a highly systematic and reliable schema for the LLM to parse and understand noisy data. It also aligns with established prompt engineering best practices for complex prompts[2]. Using this XML-structured timeline as context, we prompted the LLM to generate a cohesive, narrative-style diary of the user's previous day. For research reproducibility, we provide the raw LLM prompt text used for diary generation:

```
You are a diarist turning sensor data into a natural,
     first-person narrative.
Your goal is to create a text source for a word puzzle
     game. You must include Location Names and Points of
     Interest (POIs) exactly as written to serve as
     keywords/clues, but you must strictly adhere to the
     facts provided.

(XML input here)

STRICT CONSTRAINTS (CRITICAL)
1. STARTING THE DAY: If the log begins at 12:00 AM, do NOT
     say "I woke up." Use phrasing like "I spent the night
     at..." or "Midnight found me at..."
2. NO BIOLOGICAL ASSUMPTIONS: Do NOT assume the user is
     sleeping, hungry, or tired unless explicitly stated in
     <uEMA NOTES>.
3. AMBIENT POIS: The <POIS> list represents landmarks
     *nearby*. Do NOT say "I stopped at [POI]." DO describe
     them as surroundings (e.g., "The neighborhood features
     [POI]...").
4. TRAVEL TAG PRIORITY: If a <TRAVEL>...</TRAVEL> tag is
     present, include that travel fact exactly once in
     concise natural language. Keep origin/destination and
     distance faithful to the tag.
```

[1] https://nominatim.org/
[2] https://ai.google.dev/gemini-api/docs/prompting-strategies

```
5. ACTIVITY TAG USAGE: If <Activity> ... </Activity> tags
     are present, treat them as valid user-reported
     activities without location certainty. Include their
     content naturally in the chronology.
6. PRESERVE SPECIFIC DETAILS: You must retain the exact
     granular details, nouns, and proper names provided in
     <uEMA NOTES>. Never generalize them. For example, if a
     note says "watching the Kardashians," you must write
     "watching the Kardashians," not "watching TV." Always
     weave these exact target words smoothly and
     intuitively into the narrative.



INSTRUCTIONS
- STEP 1: CLEAN & FILTER NOTES The <uEMA NOTES> are raw
     audio transcriptions and may contain errors.
  - Filter Noise: Ignore obvious transcription errors or
     completely incoherent garble. However, do not mistake
     specific, quirky details for noise.
  - Context is King: Only include notes that make sense
     within the current location and activity context, but
     ensure their specific flavor remains intact.
- STEP 2: AGGREGATE ACTIVITIES (Crucial for Flow) Do not
     list every timestamped activity individually, but do
     not lose the specific details when summarizing.
  - Group Repeating Actions: If the notes show "Computer
     (Using)" multiple times, summarize the timeblock: *"I
     spent most of the afternoon working on the
     computer..."*
  - Smooth Transitions with Embedded Details: Combine
     related short activities while injecting the exact
     user notes. Instead of generalizing, write: *"I
     settled in for the evening and spent some time
     watching the Kardashians."*
- STEP 3: NARRATIVE GENERATION Write the entry
     chronologically using the main <TIME> tags as anchors.
  - Use the exact text inside <LOCATION>.
  - Treat <POIS> as environmental flavor/scenery. List as
     many as possible to populate the game board, but keep
     them distinct from your specific actions.
  - If <TRAVEL> exists, weave it into the narrative
     naturally (typically once near the relevant
     transition), without inventing extra trip details.
  - If <Activity> exists, include it as activity evidence
     for periods where location may be unknown.
  - Maintain a natural, diary-like flow. Avoid robotic
     lists of times within the narrative paragraphs.

OUTPUT
Write only the final diary entry. Do not write preamble.
     Start directly with the narrative.
```

D Appendix D: Full Pipeline for Puzzle Vocabulary List Generation

The **personalization module** creates and filters the daily vocabulary list for the word wheel puzzles. We designed this module to seed the word list independently of the user's data to ensure diversity of the daily puzzles. If the puzzle's underlying vocabulary were derived from users' activity data, the repetitive nature of daily routines - especially during weekdays - could rapidly introduce redundancy in the puzzle's vocabulary.

The module first queries a standard English dictionary to randomly sample a valid *seed word* containing exactly seven unique letters (e.g., "CAREFUL") to populate the word wheel. One of these seven letters is then randomly picked as the center letter, which all acceptable words must contain. Then, the module systematically extracts all words from the dictionary that are (1) at least four letters long, (2) use only the seven letters from the wheel, and (3) include the center letter. We then systematically filter the raw word list using the Python wordfreq library. All obsolete words and words with a frequency lower than $1e-7$ are dropped. Inflectional endings (e.g., "ed" and "ing") are also not used. Finally, we prompt the LLM to evaluate and prune the remaining list. The model identifies and removes all profanities, human names, proper nouns, and overly obscure jargon (e.g., specific subspecies of an animal or rare ethnic instruments) that are inappropriate for a general-audience puzzle. We have used the following exact prompt for vocabulary list pruning:

```
You will receive a list of candidate English words, each
     with a frequency and a definition. Your task is to
     decide which words to KEEP for a general-audience word
     puzzle game and which to REMOVE.

A word should be REMOVED if typical adults would probably
     not know it or would not be able to guess it easily
     from simple clues.

Remove words in any of the following categories:
1. Obsolete, archaic, or extremely rare English words. Very
     low frequency (for example < 1e-6) is a strong signal
     to remove unless the word is clearly a very common
     everyday concept.
2. Derived "-er" occupation/agent nouns that are not common
     everyday roles. Examples that should be removed
     include roles like "freer" or "tanner". Only keep -er
     words describing broad, familiar roles such as
     "teacher", "driver", "runner", "worker", "painter",
     etc.
3. Specific animal species or narrow taxonomic terms,
     especially if the definition mentions genera or says
     "one of many species". Remove terms like "tern" or any
     specific subtype of animal unless it is widely known
     to the general public.
4. Niche plant, wood, or material terms used mainly in
     crafts, manufacturing, or trade. Remove words like
     "rattan" or obscure botanical terms.
5. Esoteric religious, ceremonial, mystical, or ritual
     terms that are unlikely to be known by most people.
     Remove terms like "tantra" when they are treated as
     specialized rather than widely familiar vocabulary.
6. Old, niche, or historically specific gambling or card
     games. Remove terms like "faro".
7. Specialized food-processing, biochemical, anatomical, or
     scientific terms. Remove words like "rennet" or any
     word whose definition clearly refers to biochemical or
     anatomical components.
```

```
8. Other types of inappropriate puzzle words include:
     scientific or medical jargon, obscure geographical
     names, obscure personal names or mythological/literary
     figures, uncommon plants/fungi/minerals, dialect-only
     or regional terms, foreign loanwords not widely
     adopted in English, and crossword/Scrabble-only rare
     words.

KEEP words that are common, everyday, widely recognizable
     by typical adults, and reasonable to expect in a
     general word puzzle.
```

We added a separate, final pass for filtering person names and profanities, because the initial pruning sometimes did not successfully filter those inappropriate words. We used the following prompt as a final, sanity check:

```
You are filtering candidate words for a puzzle.

Allowed removals are ONLY these two categories:
1. Human names (first names, last names, common proper
     names used for people)
2. Profanity or sexual words inappropriate for a
     general-audience puzzle

Do NOT remove words for any other reason.
If uncertain, KEEP the word.

Input words (one per line, format: word: optional
     definition):
<<WORDS>>

Return STRICT JSON only in this exact schema:
{
  "keep": ["word1", "word2"],
  "remove": ["word3", "word4"],
  "categories": {
    "word3": "name",
    "word4": "profanity_or_sexual"
  },
  "explanations": {
    "word3": "Short reason",
    "word4": "Short reason"
  }
}
```

E Appendix E: Full Pipeline for Contextual Clue Generation and Validation

The **personalization module** bridges the puzzle vocabulary list with the user's data to author three distinct categories of clues: *Contextual, Priming, and Generic.*

- *Contextual Clues* are fill-in-the-blank style sentences that semantically anchor the puzzle vocabulary to the narrative diary context. For instance, if the LLM-generated diary mentions calling a friend on the phone during the day, and the daily puzzle contains the word *PHONE*, a sample contextual clue might read: "You were likely using your _____ to call your friend last night."
- *Priming Clues* are designed to personalize vocabulary that lacks a direct semantic link to the previous day's data, while minimizing the risk of hallucination. Given the local LLM's limited capacity, forcing associations between unrelated words and the diary context often resulted in fabricated activities. To safely bridge this gap, we pre-generated multiple-choice in-game survey questions probing users' personal traits (e.g., habits, socialization, personality) related to the target word. If a user answers this question to unlock a clue, the module incorporates their response to generate a Priming clue for the subsequent day's puzzle. For example, if tomorrow's puzzle includes the word "salad," the prior day's priming question might ask: "Do you usually eat a healthy, balanced diet?" with simple Yes/No options. A *Yes* response will yield the clue: "You maintain a healthy diet! Your daily meals very likely consist of a green _____" whereas the clue for a "No" response might instead read as: "Because you don't care much about healthy diet, you will very rarely eat a _____."
- The *Generic Clues* are crossword-style clues that function as a fallback mechanism. They are displayed only when neither contextual nor priming clues are available.

We have used the exact following prompt to generate the contextual clues:

```
CONTEXT
You are a game engine generating puzzle clues from a user's
     diary.
The user must fill in the blank to guess the word.

INPUT DATA
<Narrative diary goes here>

Word List:
<Generated and pruned puzzle vocabulary list goes here>

INSTRUCTION
Select all words that can *factually* describe the diary
     events.
Write a clue for each using one of the Accepted Framing
     Strategies.

PRIORITY: Actively scan the diary for highly specific,
     unique details (e.g., specific media watched, exact
     food eaten, unique tasks). These originate from the
     user's active reports and are the most valuable game
     content. Prioritize generating clues that incorporate
     these specific details.

ACCEPTED FRAMING STRATEGIES
1. Active Detail Integration (Top Priority): Link the word
     directly to a highly specific detail, action, or
     object explicitly mentioned in the diary.
  - *Word:* REALITY. *Diary:* User settled in to watch the
     Kardashians.
  - *Clue:* "You spent some time catching up on (blank) TV
     when you watched the Kardashians."
2. Time/Space Labeling: Use the word to label *when* or
     *where* something happened.
  - *Word:* AFTERNOON. *Diary:* User was at library at 2 PM.
```

```
  - *Clue:* "You spent a good chunk of this (blank)
      studying at <Library>."
  - *Word:* AREA. *Diary:* User went to <Neighborhood>.
  - *Clue:* "You traveled to the <Neighborhood> (blank)
      today."
3. Logical Association: Use a word that is physically
     necessary for an object/action in the diary.
  - *Word:* FARE. *Diary:* User took the subway.
  - *Clue:* "You probably paid (blank) today when taking
      the subway."
  - *Word:* EATER. *Diary:* User passed a restaurant.
  - *Clue:* "You passed <Restaurant Name>, a likely stop
      for a hungry (blank)."
4. Action Paraphrasing: Use the word to describe a movement
     mentioned in the text.
  - *Word:* REENTER. *Diary:* User went back to library.
  - *Clue:* "You went to <Library> in the morning, and then
      you (blank)-ed it after class."

BANNED STRATEGIES (Strict)
- NO IMAGINATION: Do not say "You imagined an AERATOR" or
     "You moved like an OTTER."
- NO FAKE FEELINGS: Do not say "You felt FEAR" unless the
     diary explicitly says "I was scared."
- NO CLUNKY PHRASING: The rewritten sentence must flow
     smoothly and intuitively like natural language. Do not
     force the blank into an awkward grammatical position.

OUTPUT FORMAT
Output ONLY a valid JSON list.
[
  {
    "word": "FARE",
    "clue": "You likely paid (blank) today
    when you took the subway at 11:09 AM.",
    "strategy": "Logical Association",
    "relevance_score": 5
  }
]
```

Similarly, we use the following prompt to generate priming questions, answers, and accompanying clues:

```
ROLE
You are an expert Game Designer and Behavioral
     Psychologist. You are the engine for a "Personalized
     Contextual Word Puzzle."

INPUT WORDS
<Word list here>

YOUR GOAL:
Analyze a provided list of random English words. Select
     ONLY the words that can serve as a bridge to ask a
     "Priming Question" about the user's life (habits,
     personality, health, daily routine, social status).

PROCESS FOR EACH WORD:
1. Evaluate: Can this word be the answer to a riddle based
     on a specific user trait? If no, skip it.
2. Formulate Priming Question: Create a multiple-choice
     question to gather data about the user. *Constraint:*
     Must have 2-3 concise options (e.g., "Yes/No",
     "Morning/Night", "Often/Never").
3. Generate Conditional Clues: For *each* option, write a
     clue where the target word is the answer (the
     fill-in-the-blank). The clue must explicitly reference
     the user's answer to the Priming Question.

DOMAINS OF INTEREST:
Prioritize questions that reveal:
- Social Context (Relationships, Friends, Living situation)
- Physical Health (Sleep, diet, exercise)
- Psychographics (Personality, mood, stress)
- Daily Routine (Commute, work habits, hobbies)

OUTPUT FORMAT:
Provide the response in valid JSON format.

EXAMPLE INPUT:
["stapler", "babe", "asteroid", "coffee"]

EXAMPLE OUTPUT:
{
  "selected_words": [
    {
      "target_word": "babe",
      "data_category": "Social Context",
      "priming_question": "Are you currently in a
      romantic relationship?",
      "options": ["Yes", "No"],
      "clues": {
        "Yes": "Since you mentioned you are in a
        relationship, you might affectionately call your
        partner a (blank).",
        "No": "You said you're single right now,
        so you don't have a (blank) to buy Valentine's
        gifts for yet."
      }
    },
    {
      "target_word": "coffee",
      "data_category": "Dietary Habits",
      "priming_question": "How do you usually start your
      morning?",
      "options": ["Caffeine", "Water/Nothing"],
      "clues": {
        "Caffeine": "You start your day with a jolt!
        You likely drink a hot cup of (blank).",
        "Water/Nothing": "You avoid the jitters in the
        morning, so you probably skip the cup of (blank)."
      }
    }
  ]
}
INSTRUCTIONS:
- Discard words that are too abstract or difficult to link
     to personal attributes (e.g., "asteroid").
- Ensure the tone is playful yet inquisitive.
- The clue must *always* lead the user to guess the target
     word.
```

Finally, we generate the crossword-style generic fallback clues using the following prompt:

```
ROLE
You are a Crossword Puzzle Constructor.
Your task is to write fun, easy-to-medium difficulty clues
    for a list of words.

INPUT WORDS
<Word list goes here>

INSTRUCTIONS
For each word in the list, write one clue.
1. Style: Mix it up! Use definitions, synonyms, antonyms,
    or common phrases (fill-in-the-blank).
2. Difficulty: Aim for "Monday Morning Crossword"
    difficulty. Accessible but clever.
3. Constraint: Do NOT use the word itself (or variations of
    it) in the clue.
  - *Bad:* "To run fast." (Target: RUNNING)
  - *Good:* "Participating in a marathon."(Target: RUNNING)

OUTPUT FORMAT
Output ONLY a JSON list of objects:
[
  {
    "word": "TARGET",
    "clue": "Your clue here."
  }
]
```

Finally, the LLM executes an additional validation pass over all generated clues, assigning each contextual clue a score between one to five based on its correctness and quality. We designed this additional QA pass to identify and drop any clues containing logical stretches, fabricated events, or hallucinations that are not grounded in the diary context. We only keep the clues which have an assigned score of four or higher, which are included in the finalized puzzle JSON. We use the exact following prompt:

```
ROLE
You are a Quality Control bot for a diary puzzle.

INPUT DATA
Diary: <Narrative diary goes here>

Clues: <Generated contextual clues go here>

TASK
Rate each clue (1-5) based on Contextual Validity.

SCORE 5 (VALID):
- Fact: "You passed Caffe Nero." (NERO) -> 5
- Definition: "You went to <Neighborhood>, which is an
    AREA." -> 5 (True definition).
- Logical Link: "You took the subway, so you paid FARE." ->
    5 (Logical assumption).
- Time/Place: "You were at the library in the AFTERNOON."
    -> 5 (True time).

SCORE 1 (INVALID/HALLUCINATION):
- Creative Writing: "You felt light as an OTTER." -> 1
    (Hallucination).
- Fake Objects: "You imagined an AERATOR." -> 1 (Not in
    diary).
- Fake Actions: "You ate an ENTREE." -> 1 (Diary only says
    'passed restaurant').


OUTPUT FORMAT
Output ONLY a valid JSON list.
[
  {
    "word": "TARGETWORD",
    "clue": "Original clue...",
    "score": 5,
    "reason": "Valid logical link. Subway implies Fare."
  }
]
```

F Appendix F: Usability Metrics and Burden Survey

Distribution of individual NASA-TLX scores for the smartwatch μEMA system and the word wheel puzzle games are displayed in Figure 9. System usability scale (SUS) scores for the smartwatch prompting system and the puzzle games are displayed in Figure 10. Finally, the distribution of burden survey responses for individual items asking about experience with puzzle games are displayed in Figure 11.

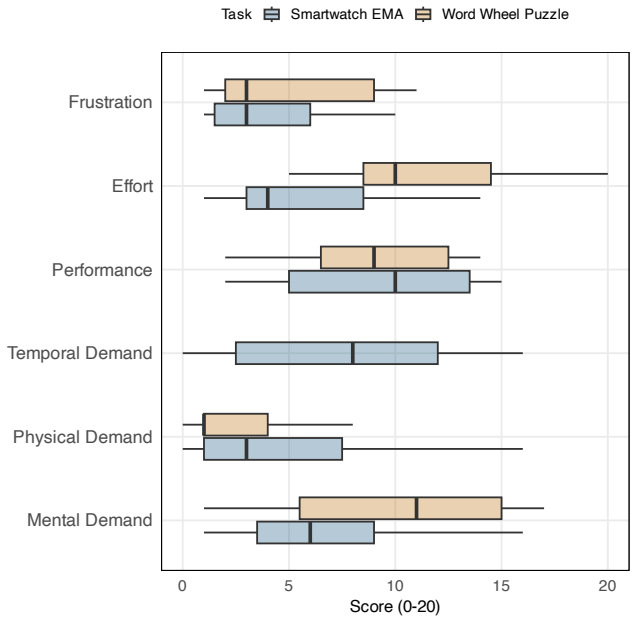


Figure 9: Distribution of NASA-TLX subscale ratings for the smartwatch prompting system and the word wheel puzzle game.

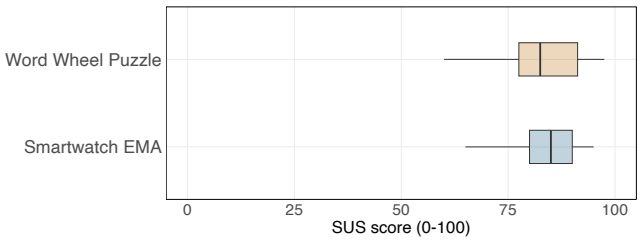


Figure 10: Distribution of SUS scores for the smartwatch prompting system and the word wheel puzzle game.

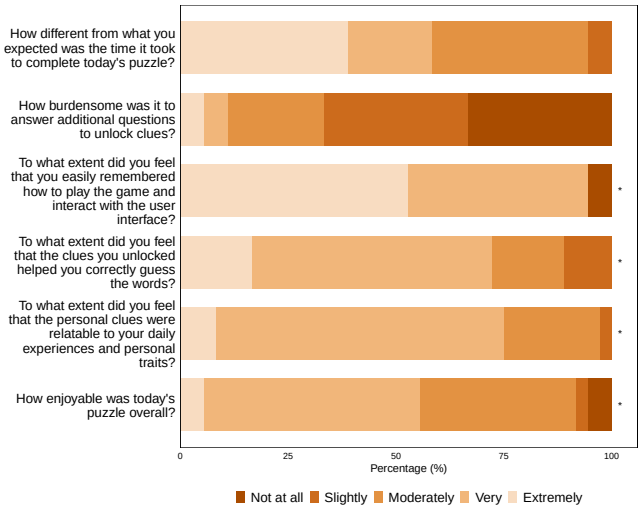


Figure 11: Distribution of responses for individual items related to puzzle games in the daily burden survey. Survey items with an asterisk (*) are reverse coded.

G Appendix G: Categories of Semantic Tags and Additional Examples

Semantic tags are distinct pieces of contextual information about the participant, such as what the person is doing, in what posture, when, and using what. We identified and counted semantic tags, because traditional NLP metrics such as word counts or preposition counts cannot reliably capture how much additional details participants are providing in their verbal self-reports. For example, the sentences "I am sitting and just working" and "Sitting and working on programming assignment" have the same word counts but the second phrase carries richer contextual information than the first phrase. We have classified the semantic tags in participants' μEMA self-reports into the following categories:

- **Posture:** Refers to how people naturally orient and hold their bodies. Examples include: *Upright, Sitting, Standing, and Lying.*
- **Activity:** Refers to the primary action that a person is currently doing. Examples include *Walking, Cooking, Talking, and Grocery Shopping.*
- **Objects:** Refers to physical items used as tools to support a specific activity. To keep our categories mutually exclusive, we only coded an item as an *Object* if **it was attached to a clear, separate action.** For example, if a participant reported "making presentation slides on a laptop," *making presentation slides* was coded as the Activity and *laptop* as the Object. However, if they reported a vague phrase where the item was the whole action—such as "using laptop"—we simply coded the entire phrase as an Activity. We manually reviewed these edge cases to prevent any tag from being counted twice.
- **Location:** Refers to a location where a person is currently at. Examples include *In my room, In classroom, In the subway, In the bus, At college.*
- **Time:** Refers to any mention of *when* an activity happens. This can be an exact time of day, or simply stating that an event just finished or is about to start. Examples include *Just woke up, About to go home, and Meeting at 7 PM.*
- **Social:** Refers to any mentions of social contexts. Examples include *With a friend, In a group meeting, and With my wife.*
- **Purpose:** Refers to a person's intention of performing an action. Similar to objects we only coded an item as a *Purpose* if it was attached to a clear, separate action. Examples include *Walking to campus "to attend a class", Opening a pantry "to get medicines", and Walking to subway station "to go to work."*
- **State:** Refers to any mentions of self-reported physical or emotional states. Examples include *I'm tired, I'm hungry, and I'm sleepy.*
- **Specificity:** Refers to extra details that describe an activity or item in more depth. For example, in the phrase "*Watching a movie on Italian food culture,*" *Italian food culture* is coded as *Specificity* because it simply adds context to "*Watching a movie.*"

For reproducibility, we provide the exact LLM prompt text used for semantic parsing below:

```
You are an expert qualitative data coder. Your task is to
     analyze ecological momentary assessment (EMA)
     self-reports and score their "semantic richness" by
     extracting and counting "atomic information units".

DEFINITIONS & CATEGORIES

An atomic information unit is one distinct, non-redundant
     piece of contextual information.
- POSTURE: Body position (sitting, standing, lying down,
     walking).
- What they are doing (working, eating breakfast, reading,
     cleaning, watching).
- SPECIFICITY: Narrows a generic activity. *Only counts if
     the broader activity is also stated.* (e.g., "spelling
     bee" narrows "puzzle").
- PURPOSE: Why they are doing it (for a class, to relax,
     for work).
- OBJECT/TOOL: Instrument used OR the specific target of
     the action (laptop, phone, counters). *Only counts if
     it adds info beyond the activity.*
- LOCATION: Where they are (bus, desk, library, outside,
     floor, chair, bed).
```

```
- SOCIAL: Social context (alone, with friends, in a
    meeting).
- TEMPORAL: Time context (just woke up, during lunch break,
    at 7).
- AFFECT: Emotional state (stressed, bored, enjoying it).

STRICT RULES
1. Atomic: One piece of explicit information = one unit.
2. Collapse Redundancy: "Working on code" and "writing
    code" in the same report = 1 ACTIVITY unit.
3. Specificity Threshold: "Typing code" = 1 ACTIVITY.
    "Completing an assignment writing code" = working
    (ACTIVITY) + assignment (PURPOSE) + code (SPECIFICITY)
    = 3 units.
4. No Inference & No Hallucination: Count ONLY what is
    explicitly stated in the text. Do not invent words
    (e.g., if they say "filling", do not output
    "assignment").
5. The Furniture Rule: Surfaces and furniture (floor, desk,
    chair, bed) are ALWAYS a LOCATION, NEVER a POSTURE.
6. Do Not Drop Data: You must extract every distinct
    activity (e.g., if they are eating AND watching,
    extract both) and every social contact (friends,
    colleagues).

COMMON EDGE CASES (DO NOT MISS THESE)
- Idiomatic Phrasing: Treat phrases like "sitting on my
    laptop" or "on my phone" as POSTURE (sitting) + OBJECT
    (laptop/phone). Do not classify the device as an
    activity or location.
- Meals: "Eating breakfast/lunch" is an ACTIVITY, not a
    TEMPORAL unit.
- Specific Targets: You must extract specific environmental
    targets (e.g., "cleaning the counters" = ACTIVITY
    (cleaning) + OBJECT (counters)).

EXAMPLES
- Input: "Sitting down working on my desk, completing an
    assignment writing code."; Output: sitting (POSTURE),
    at desk (LOCATION), working/writing code (ACTIVITY),
    completing an assignment (PURPOSE)  4
- Input: "Sitting on the floor, eating breakfast and
    watching on the laptop."; Output: sitting (POSTURE),
    on floor (LOCATION), eating breakfast (ACTIVITY),
    watching (ACTIVITY), laptop (OBJECT)        5
- Input: "I'm sitting on my laptop, doing some work.
    Helping a friend out with our assignment."; Output:
    sitting (POSTURE), laptop (OBJECT), doing work
    (ACTIVITY), helping a friend (SOCIAL), with assignment
    (PURPOSE)        5

TASK
Process the following batch of reports. Output exactly one
    line per report using this strict tab-separated format
    (do not include headers, markdown tables, or
    conversational text):
[Extracted units with categories]
[Integer Total Count]

INPUT BATCH:
[Insert 5 Rows Here]
```